Shared Errors Limit the Benefits of Multi-Agent Collaboration in Medical Vision-Language Models

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Abstract

Multi-agent systems (MAS) built from one vision-language model (VLM) assume that role-prompted agents provide partly independent readings. Agents that share weights can agree because they make the same error. We test this assumption with four proprietary and five open-weight VLMs on VQA-RAD, SLAKE and MedXpertQA-MM. We measure error correlation and the effective number of independent agents, then probe shared errors using blank images, region occlusion, paired images and query-agnostic reports that separate perception from reasoning. Nine agents of one VLM are worth 1.17 *to* 1.99 *independent agents (N_{eff}), and a beta-binomial model predicts majority accuracy from mean agent accuracy and error correlation. On MedXpertQA-MM,* 64.2% *to* 76.2% *of shared wrong answers match the agents' blank-image answer. Sharing one report raises error correlation; changing the model lowers it by* 0.470 *to* 0.492*, the largest reduction among the tested contrasts. Agents change correct minority answers to wrong answers during debate. Code and data will be released.*

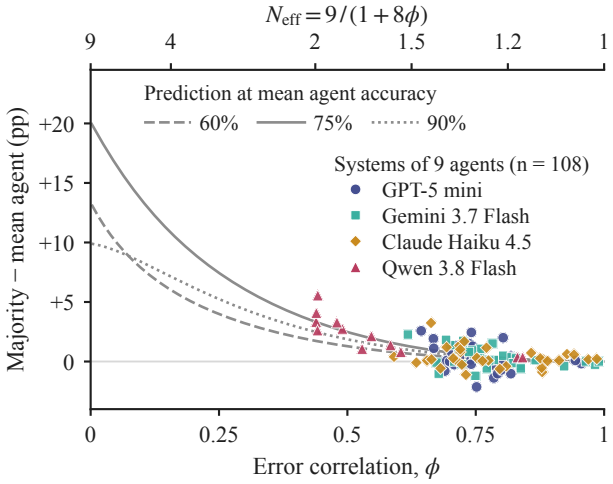


Figure 1. **The benefit of majority voting depends on error correlation.** Majority minus mean agent accuracy versus ϕ for 108 systems of 9 agents on yes/no items. Gray curves: beta-binomial predictions at mean agent accuracy 60% (dashed), 75% (solid), and 90% (dotted). Top axis: $N_{\text{eff}} = 9/(1+8\phi)$. Blue circles: GPT-5 mini; teal squares: Gemini 3.7 Flash; amber diamonds: Claude Haiku 4.5; rose triangles: Qwen 3.8 Flash.

1. Introduction

Double reading pays off when the second reader errs on different cases from the first reader [8, 16, 26]. Medical multi-agent systems coordinate role-prompted instances of one model through voting or multiagent debate [7, 13, 14, 24].

Agents of one VLM share the vision encoder, the language model and its language prior. A role prompt changes instructions; it changes neither what the vision encoder sees nor the answer the model falls back on when the image does not decide the question. Majority vote corrects an agent when others err on different items [5]; when they fail together, it returns their shared error. The gain available to a MAS of one VLM is set by error correlation rather than agent count.

We measure error correlation ϕ and the effective number of independent agents N_{eff} [15], and relate them to majority accuracy with a beta-binomial model (Fig. 1). We vary the image through blanking, reduced resolution, occlusion and pairing, and vary shared components through roles, contrast adjustment, in-context examples, reports and models. Query-agnostic reports separate perception from reasoning [21]. We compare Independent and Decentralized MAS with single-agent and self-consistency baselines across proprietary and open-weight VLMs (Sec. 3).

Our contributions are:

- **Measuring dependence.** We measure how many independent agents a MAS of one VLM provides and show that mean agent accuracy and error correlation predict majority accuracy. We decompose multi-agent gains into gains from better agents and gains from the vote.
- **Tracing shared errors.** We use blank-image probes to

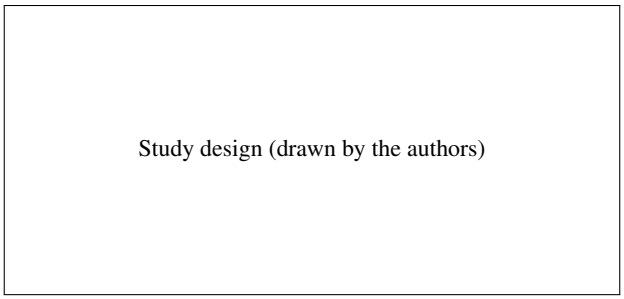


Figure 2. **Controlled comparisons vary visual evidence and shared components.** Systems are single agent, self-consistency, Independent and Decentralized; image manipulations include blank images, reduced resolution, region occlusion and paired images. Query-agnostic reports separate perception from reasoning; outcomes are accuracy, ϕ and N_{eff}.

show that shared wrong answers follow the answer the model gives without visual evidence.

- **Separating shared components.** We use query-agnostic reports to show that the perception module sets accuracy and a shared report sets error correlation. Matched contrasts show that changing the model decorrelates agents more than changing their role or input.
- **Tracking conformity.** We compare initial with final answers to show that debate changes correct minority answers to wrong answers.

2. Related work

Multi-agent systems in medicine. Role-prompted medical MAS improve medical question answering [13, 24], and their gains shrink as the base model becomes more capable [14]. Multiagent debate supports revision from peer responses [7]; benchmarks and debate analyses characterize collaboration and diversity [33, 34]. We connect these systems' radiology accuracy to their effective number of independent agents and identify shared components behind correlated errors.

Correlated errors and aggregation. Dependence limits information pooling [5], and correlated wrong answers occur across language models [12]. Kish's design effect relates correlation to effective sample size [15]; diagnostic pooling benefits from independent judgments with similar accuracy [16]. We predict majority accuracy from mean agent accuracy and error correlation, then measure changes in dependence through matched component contrasts.

Language priors and visual reliance in VLMs. Visual-question-answering benchmarks show that language priors affect answers [4, 9] and that models can fail on paired images requiring different answers [18, 27]. Medical probes expose diagnosis failures and flawed image comprehension behind correct answers [11, 30]. Counterfactual grounding tests visual reliance [31], while Prism separates perception from reasoning through visual descriptions [21]. We trace shared MAS errors through blank-image answers and report swaps that isolate dependence on common findings.

Table 1. **Probes and coverage.** Main denotes VQA-RAD, SLAKE and MedXpertQA-MM (250 items each); SLAKE-pairs has 200 pairs. The 4 proprietary models are GPT-5 mini, Gemini 3.7 Flash, Claude Haiku 4.5 and Qwen 3.8 Flash; where 3 are listed, Qwen is excluded.

Probe	Data sets	Conditions	Models
Original images	Main	$\leq$1024 pixels	4 proprietary
Reduced resolution, blank images	Main	256 and 128 pixels; uniform gray	4 proprietary
Region occlusion	SLAKE-target	Target or control box	3 proprietary
Paired images	SLAKE-pairs	One question, two images, opposite answers	3 proprietary
Query-agnostic reports	Main	Own or shared report	3 perception $\times$ 3 reasoning modules
Matched contrasts	Main	Role, contrast adjustment, model; in-context examples (VQA-RAD, SLAKE)	4 proprietary (in-context examples: 3)
Open-weight probes	All five	Original, reduced-resolution and blank images; occlusion; paired images	5 open-weight

3. Experimental setup

3.1. Agents and multi-agent systems

A MAS comprises 9 instances of one VLM (Fig. 2). The single agent uses a general-radiologist prompt; self-consistency samples that prompt with majority vote [29]. Independent MAS agents receive subspecialty roles, answer separately and vote. The Decentralized MAS debates for at most 3 rounds from the same initial answers and votes over final answers [7, 14].

3.2. Error correlation and majority accuracy

Error correlation ϕ is the mean pairwise Pearson correlation of binary error indicators across items; pairs with constant indicators are omitted. The effective number of independent agents is $N_{\text{eff}} = 9/(1 + 8\phi)$ [15]. On yes/no items, a beta-binomial model predicts majority accuracy from mean agent accuracy $\bar{p}$ and intraclass correlation ϕ:

$$\alpha = \frac{\bar{p}(1-\phi)}{\phi}, \qquad \beta = \frac{(1-\bar{p})(1-\phi)}{\phi}, \tag{1}$$

$$K \sim \text{BetaBin}(9, \alpha, \beta), \quad P(\text{majority correct}) = P(K \geq 5). \tag{2}$$

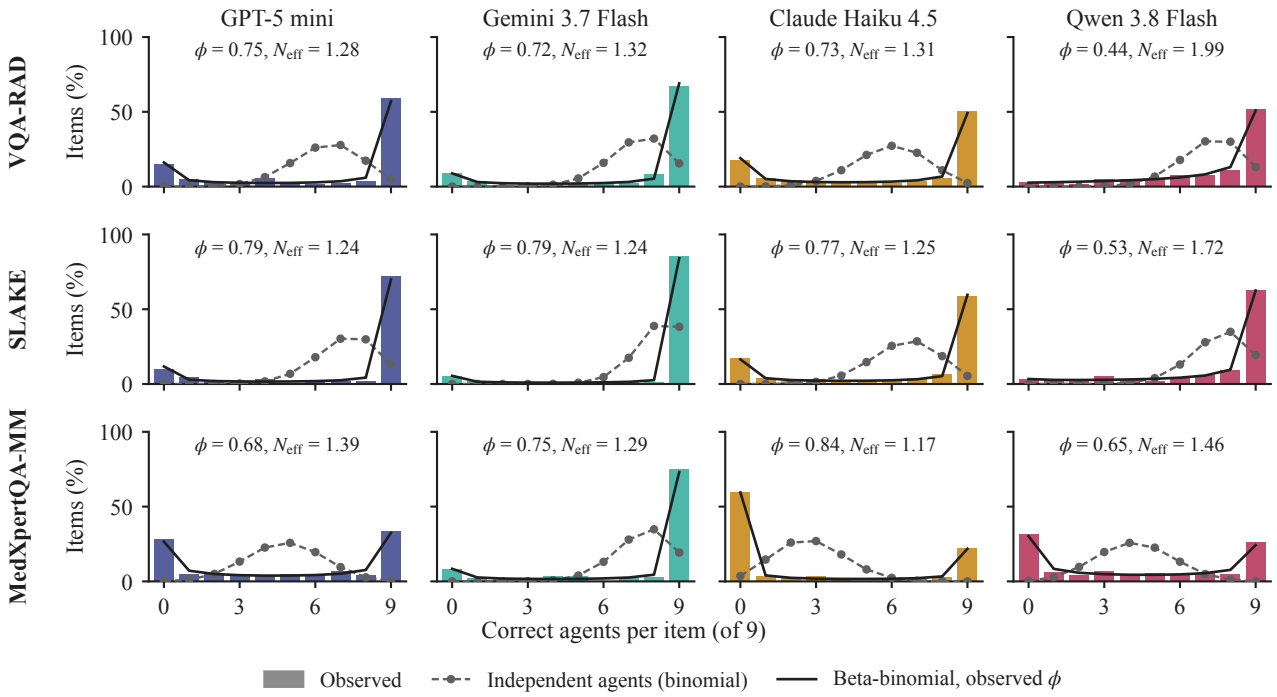


Figure 3. **Agents of one VLM are right or wrong together.** Rows show VQA-RAD, SLAKE and MedXpertQA-MM (250 items each); columns show GPT-5 mini, Gemini 3.7 Flash, Claude Haiku 4.5 and Qwen 3.8 Flash. Bars ("Observed") show the percentage of items with 0 to 9 correct agents in the Independent MAS with original images. Dashed lines with dots ("Independent agents (binomial)") show the binomial distribution for 9 independent agents with the same mean agent accuracy; solid lines ("Beta-binomial, observed ϕ") show the beta-binomial distribution with the observed error correlation. Each plot reports ϕ and N_{eff}.

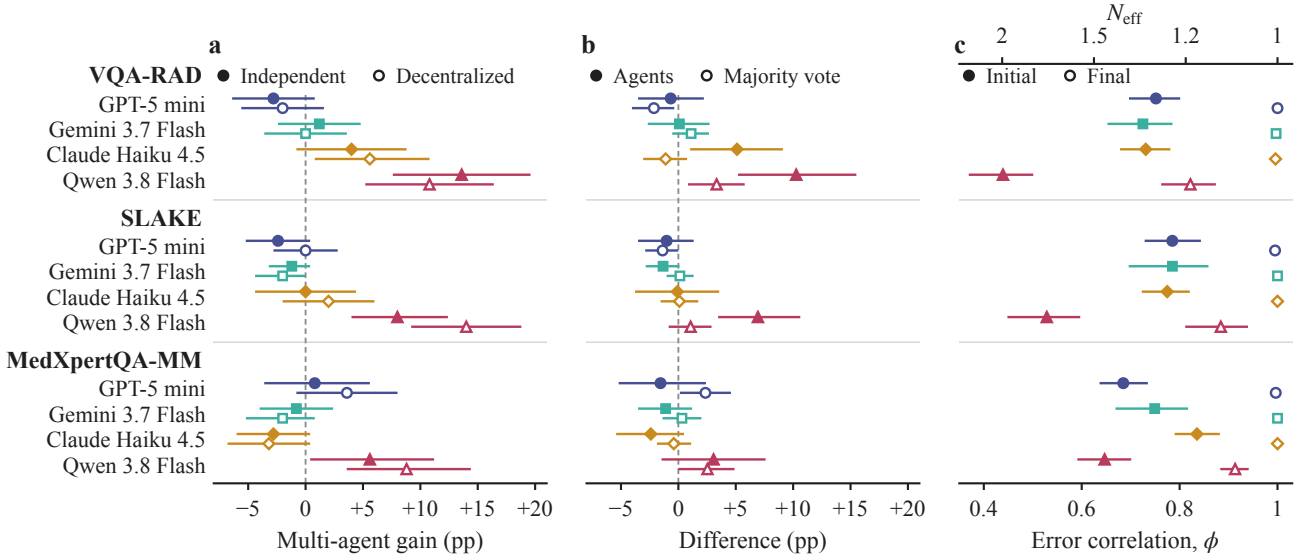


Figure 4. **Nine agents from one VLM are worth fewer than two.** Rows group models within data sets (250 items each). (a) Multi-agent gain: Independent (filled) and Decentralized (open) accuracy minus single-agent accuracy. (b) Independent gain decomposed into mean agent minus single-agent accuracy (Agents, filled) and majority minus mean agent accuracy (Majority vote, open). (c) Error correlation ϕ for initial (filled) and final (open) answers; the top axis gives N_{eff}. Model colors as in Fig. 1.

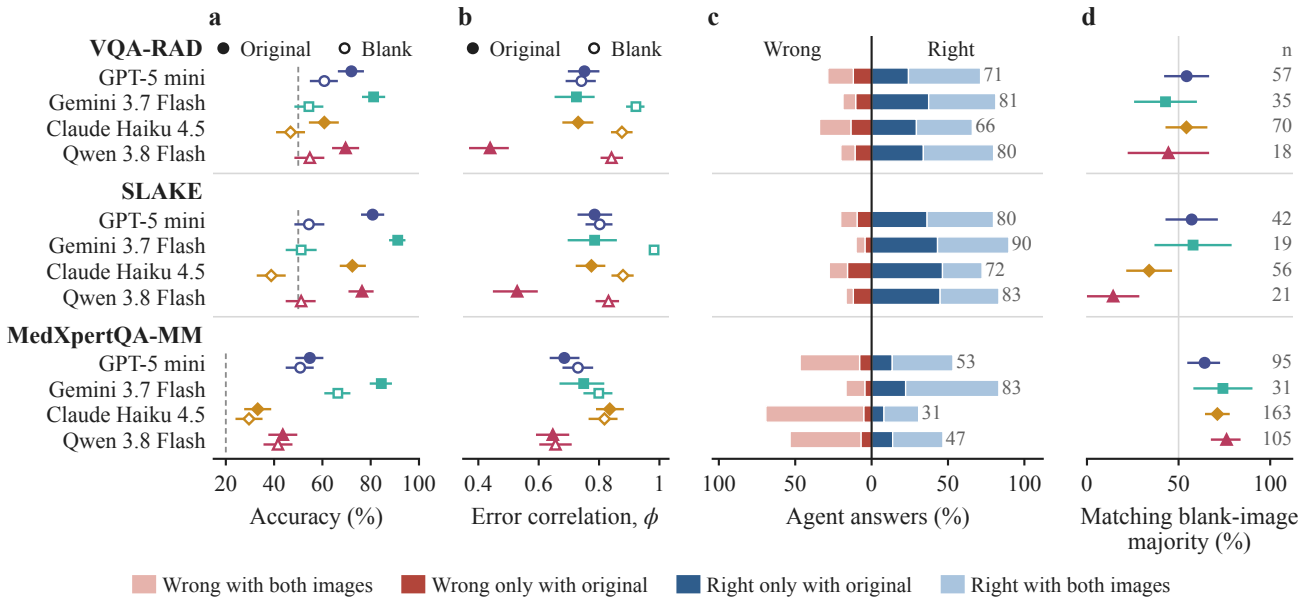


Figure 5. **Most shared wrong answers on MedXpertQA-MM match the blank-image majority.** (a) Single-agent accuracy with original (filled) and blank (open) images; dashed lines mark chance. (b) Independent MAS error correlation with each image. (c) Agent answers cross-tabulated by correctness: wrong with both images, wrong only with original, right only with original, and right with both images; end numbers give original-image accuracy. (d) Shared wrong answers whose majority matches the blank-image majority; n counts items with at least 7 of 9 agents wrong. Each data set has 250 items. Model colors as in Fig. 1.

Multi-agent gain (MAS minus single-agent accuracy) splits into mean agent minus single-agent accuracy plus majority-vote minus mean agent accuracy.

3.3. Probes

Image probes use uniform gray blank images, reduced resolution, target or control occlusion, and paired images with opposite reference answers (Tab. 1). Occlusion sensitivity is target minus control answer-change rate [32]; pair accuracy requires both answers correct. Shared wrong answers have at least 7 of 9 agents wrong. Perception modules write query-agnostic reports without questions or options; reasoning modules answer from their own or a shared report without the image [21]. Matched contrasts compare ϕ without versus with a shared component: role, contrast adjustment ($\gamma = 0.7$ or CLAHE [35]), in-context examples (3 shared or private), report, perception module, reasoning module or model. Conformity is the share of initially correct answers that become wrong after debate [1, 33].

3.4. Data and models

VQA-RAD and SLAKE use closed yes/no questions [17, 19]; MedXpertQA-MM retains radiologic images [36]. SLAKE-target has annotated target and control regions; SLAKE-pairs uses opposite-answer images [18, 27]. Proprietary-model coverage appears in Tab. 1. Open-weight

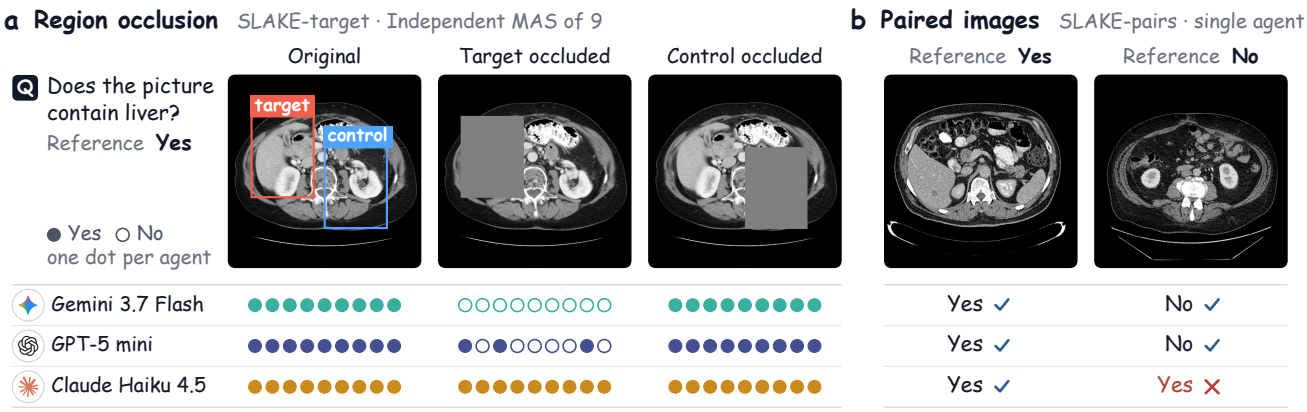


Figure 6. **Occluding the queried organ changes Gemini agents' answers, while Claude agents keep theirs.** (a) Region occlusion (SLAKE-target): "Does the picture contain liver?" (reference Yes). Rows identify Gemini 3.7 Flash, GPT-5 mini and Claude Haiku 4.5; columns show the original image (target box red, control box blue), target occluded and control occluded, as sent to the models. Each cell shows 9 agents of the model's Independent MAS: filled dots denote Yes, open dots No. (b) Paired images (SLAKE-pairs): the same question on images with reference Yes and No; one single-agent answer per model, with a check for correct and a cross for wrong.

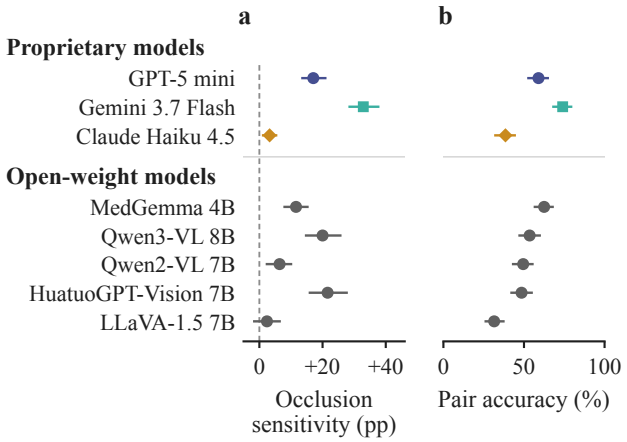


Figure 7. **Models differ in their sensitivity to the queried image region.** (a) Occlusion sensitivity, target minus control answer-change rate, on SLAKE-target (250 items). (b) Pair accuracy, both answers correct, on SLAKE-pairs (200 pairs with opposite reference answers). Occlusion estimates average Independent MAS agents for proprietary models; pair accuracy uses single agents. Open-weight models use one forward pass per item. Model colors as in Fig. 1.

models are MedGemma 4B, Qwen3-VL 8B, Qwen2-VL 7B, HuatuoGPT-Vision 7B and LLaVA-1.5 7B [2, 3, 20, 23, 28].

3.5. Statistics

Abstentions (unparsed answers) count as wrong; details are in Sec. S1.

4. Results

4.1. Nine agents are worth fewer than two

In every system, most items were answered correctly by all nine agents or by none, a U-shaped distribution that the beta-binomial with the observed ϕ reproduces and the binomial for independent agents does not (Fig. 3). The Independent MAS provided $N_{\text{eff}} = 1.17$ to 1.39 for GPT-5 mini, Gemini 3.7 Flash and Claude Haiku 4.5, and 1.46 to 1.99 for Qwen 3.8 Flash (Fig. 4). For GPT-5 mini, Gemini and Claude, Independent and Decentralized multi-agent gains stayed within -3.2 to 5.6 percentage points (pp) (Fig. 4a). Qwen's gains arose before the vote: mean agent minus single-agent accuracy was 3.1 to 10.3 pp, while majority minus mean agent accuracy was 1.1 to 3.3 pp (Fig. 4b). Observed majority accuracy differed from the beta-binomial prediction by -0.07 pp on average (SD 0.84 pp) across 108 yes/no systems (Fig. 1). Debate drove final-answer ϕ to 0.995 to 1.000 for GPT-5 mini, Gemini and Claude (Fig. 4c; Tab. S2).

4.2. Shared wrong answers follow the language prior

On MedXpertQA-MM, 64.2% to 76.2% of shared wrong answers matched the blank-image majority (Fig. 5d; Tab. S4). Blank images cost single agents 11.2 to 40.0 pp on VQA-RAD and SLAKE, versus 2.0 to 4.0 pp for GPT-5 mini, Claude and Qwen on MedXpertQA-MM and 18.0 pp for Gemini (Fig. 5a). With blank images, Gemini, Claude and Qwen reached $\phi = 0.831$ to 0.982 on VQA-RAD and SLAKE, up from 0.439 to 0.785 with original images (Fig. 5b; Tab. S3).

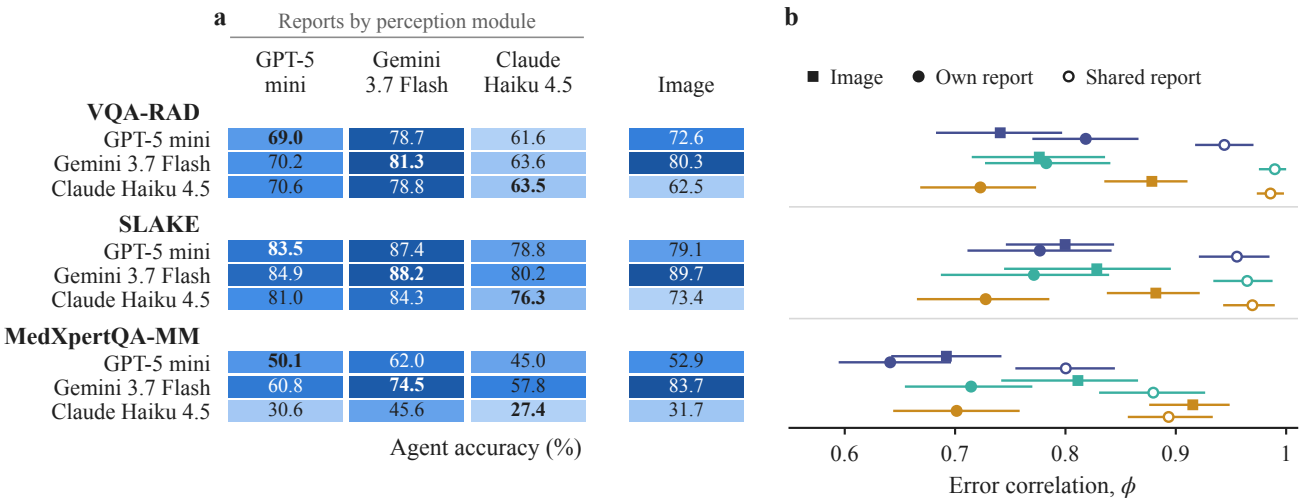


Figure 8. **Perception sets accuracy, while sharing a report increases error correlation.** (a) Agent accuracy with its own query-agnostic report from the column's perception module, or with the image. Rows identify reasoning modules within data sets (250 items each); shading scales within each data set. Bold cells use the same model for both modules. (b) Error correlation among 9 agents: image (filled square), own report (filled circle), and shared report (open circle), with same-model reports. Model colors as in Fig. 1.

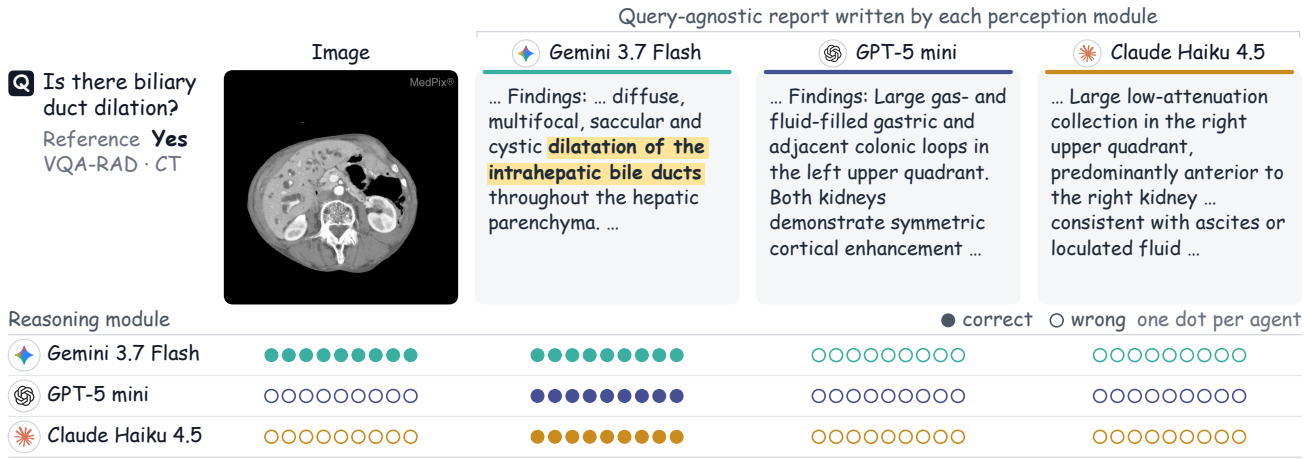


Figure 9. **The perception module determines whether reasoning agents can identify biliary duct dilation in this case.** VQA-RAD CT: "Is there biliary duct dilation?" (reference Yes). Columns show the image, then verbatim excerpts of the first query-agnostic report from Gemini 3.7 Flash, GPT-5 mini and Claude Haiku 4.5; the finding that answers the question is highlighted. Rows identify reasoning modules. Each cell shows 9 agents: self-consistency agents reading the image, or agents each reading its own report from the column's perception module. Filled dots denote correct answers, open dots wrong answers.

4.3. Models differ in where they look

Gemini's occlusion sensitivity was 32.9 pp, versus 17.1 pp for GPT-5 mini and 3.2 pp for Claude (Fig. 7a). Pair accuracy was 74.0% for Gemini, 59.0% for GPT-5 mini and 38.5% for Claude (Fig. 7b; Tab. S5). Occluding the liver changed every Gemini agent's answer and no Claude agent's answer; Claude answered yes to both paired images (Fig. 6).

4.4. Perception sets accuracy; a shared report sets correlation

Gemini reports raised the other reasoning modules' accuracy by 6.1 to 16.3 pp over reading the image (Fig. 8). With identical Gemini reports on MedXpertQA-MM, reasoning modules reached 74.5% (Gemini), 62.0% (GPT-5 mini) and 45.6% (Claude). Sharing one report raised ϕ to 0.800 to 0.990, from 0.641 to 0.818 with own reports from the same model (Fig. 8b; Tab. S7). Only Gemini's report mentioned

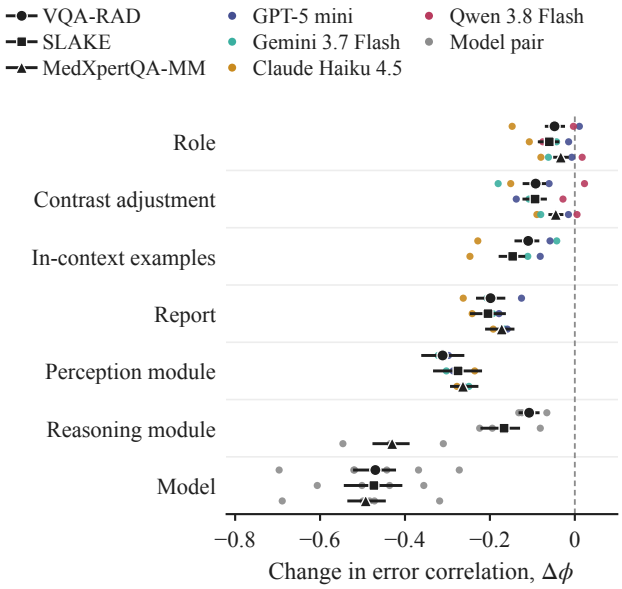


Figure 10. **Changing the model reduces error correlation more than changing a component within one model.** Matched $\Delta\phi$ compares agents differing in the named component with agents sharing it. Black markers give data-set means: circle, VQA-RAD; square, SLAKE; triangle, MedXpertQA-MM (250 items each). Colored dots give models; gray dots give model pairs. In-context examples compare private with shared examples on VQA-RAD and SLAKE. Report compares own with shared reports; reasoning-module contrasts hold the report fixed. Model colors as in Fig. 1.

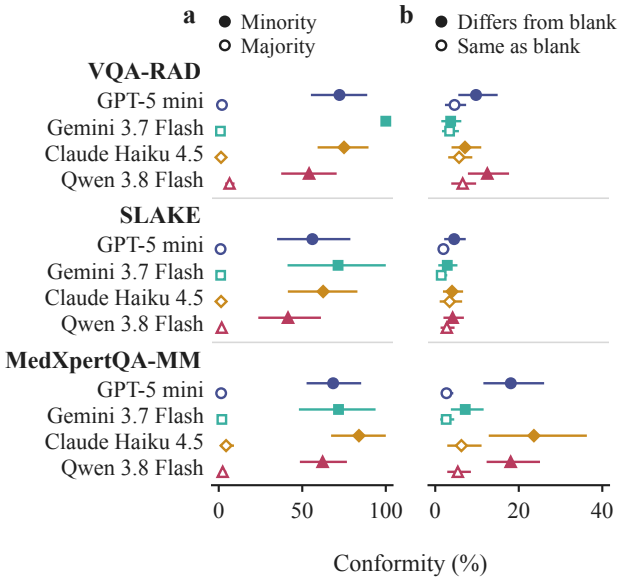


Figure 11. **Agents change correct minority answers to wrong answers during debate.** Conformity is the share of initially correct agents whose final answer is wrong in the Decentralized MAS. (a) Initial minority (filled) versus majority (open). (b) Initial answer differs from (filled) or matches (open) the same agent's blank-image answer. Rows group models within data sets (250 items each; 9 agents). Model colors as in Fig. 1.

the dilated bile ducts in Fig. 9; every reasoning module answered correctly from it and none from the other reports.

4.5. Changing the model decorrelates errors most

Self-consistency agents of one model already make correlated errors, ϕ 0.692 to 0.915 for GPT-5 mini, Gemini and Claude and 0.442 to 0.629 for Qwen (Tab. S8). Changing the model lowered ϕ by 0.470 to 0.492, the largest reduction among tested contrasts (Fig. 10). Changing the role produced the smallest reduction, 0.033 to 0.060; contrast adjustment, in-context examples, report and perception module lay between. On MedXpertQA-MM, changing the reasoning module under an identical report lowered ϕ by 0.430, more than changing the perception module (0.263; Tab. S8).

4.6. Debate removes correct minority answers

Debate changed 41.5% to 100.0% of correct initial minority answers to wrong answers, versus 1.0% to 6.5% of correct majority answers (Fig. 11). On MedXpertQA-MM, conformity was 7.2% to 23.6% for correct answers that differed from the same agent's blank-image answer, versus 2.6% to 6.3% for those that matched (Tab. S6).

5. Discussion

The return from adding agents of one VLM is predictable from mean agent accuracy and ϕ (Fig. 1). A MAS design can be tested by its effect on these quantities. The gain decomposition distinguishes improved agents from a better vote, as for Qwen (Fig. 4).

Shared errors reflect common perception and language priors when images do not decide the answer (Figs. 5 and 8). Matched contrasts rank changes to perception and to the answering model above roles or contrast adjustment (Fig. 10). These contrasts predict that prompt diversity within one VLM cannot produce independent errors.

Debate trades disagreement for unanimity (Figs. 4 and 11). Correct minority answers become wrong, especially on MedXpertQA-MM when they differ from the agent's blank-image answer. Keeping initial answers separates errors that communication corrects from those it creates.

An AI second reading in radiology is measured by its joint error with the first reading [8]. Our contrasts predict greater complementarity from a different model than another instance of the same VLM (Fig. 10). Reader studies

can test this prediction alongside the influence of wrong AI suggestions on radiologists [6].

Limitations. The evaluation uses closed questions from public data sets and includes no human readers or clinical reporting.

6. Conclusion

Agents of one VLM share perception and the language prior, so they fail on the same items and majority vote and debate recover little. Error correlation, rather than the number of agents, sets what a MAS can gain. A second reading decorrelates errors when it comes from a different model, not a different prompt.

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Shared Errors Limit the Benefits of Multi-Agent Collaboration in Medical Vision-Language Models

Supplementary Material

S1. Protocol details

S1.1. Exclusions and image preparation

Image eligibility was established before sampling. VQA-RAD used closed yes/no questions from its test split; SLAKE used closed yes/no questions from its English test split. Duplicate image–question combinations were removed from VQA-RAD. For MedXpertQA-MM, GPT-4.1 nano labeled image modality, and Claude Haiku 4.5 also labeled each candidate image as radiologic or not. Drawings and diagrams were excluded, as were photographs, pathology slides, charts and tables. Every attached image had to be radiologic for an item to be eligible; these questions had 5 answer options.

SLAKE-target required an annotated organ or lesion that was present in the image. It used the English training and test splits and excluded items in the SLAKE evaluation set. The target bounding box was expanded by 10% on each side and clipped to the image boundaries; the expanded box occupied at most 40% of the image. An equal-sized control box did not overlap the target and had at least 70% of its area inside the body. Target and control occlusion used mid-gray fills before resizing. SLAKE-pairs contained the same question with opposite reference answers and allowed at most 5 pairs per question template.

Sampling used stratified shuffling with seed 42. VQA-RAD strata used the reference answer; SLAKE strata also used modality. MedXpertQA-MM strata used body system and medical task. SLAKE-target strata used target type and modality; SLAKE-pairs strata used question template. Yes/no option order was randomized separately for each item from its identifier. Images were resized proportionally, without upsampling, to at most 1024 pixels on the long side. Reduced-resolution conditions capped that side at 256 or 128 pixels. Blank images were uniform gray images of the same dimensions [31].

S1.2. Agent prompts and sampling

Role assignment used the question text to select subspecialties. GPT-4.1 nano selected and ranked the 9 most relevant roles from a roster of 12. The roster covered neuroradiology, head and neck, thoracic, cardiovascular, abdominal, genitourinary, musculoskeletal, breast, pediatric, nuclear medicine, interventional and emergency radiology. If fewer than 9 roles were returned, the remaining positions were filled in roster order. Systems of 1, 3 and 9 agents used the first members of the selected roles and reused their initial answers. The single agent and self-consistency baseline used a general-radiologist prompt, with instructions to reason briefly, step by step, before selecting an answer.

Generation settings were fixed by model and condition. The configuration seed was 1; agent i used seed $100 + i$. Where supported, agents in a MAS, debate rounds and perception modules used temperature 0.7; the single agent used 0.3. GPT-5 mini accepted neither seed nor temperature and used low reasoning effort, with separate sampling for its agents. GPT-4.1 nano used temperature 0 for role assignment and modality screening. Decentralized MAS reused Independent MAS initial answers and allowed up to 3 subsequent rounds, with a stop at unanimity. Agents saw peer answers and reasoning; peer context was capped at 900 characters and peer order was shuffled by item and round.

S1.3. Contrast adjustment

Contrast adjustment assigned different image versions to self-consistency samples. Agents 1–3 saw the original image; agents 4–6 saw a gamma-transformed image with gamma 0.7. Agents 7–9 saw contrast-limited adaptive histogram equalization (CLAHE) [35]. CLAHE used a clip limit of 2.0 and an 8×8 tile grid on the lightness channel. The corresponding contrast compared agents with different adjustments against self-consistency samples with the original image.

S1.4. In-context example retrieval and selection

In-context examples came from training images with closed yes/no question–answer pairs. This condition used VQA-RAD and English SLAKE training data and was evaluated on VQA-RAD and SLAKE. The data sets carried no patient identifiers, so exclusions used image identity and similarity. SHA-256 hashes excluded images identical to any test image. For each query, candidates were also excluded within Hamming distance 8 of its 64-bit perceptual hash or at CLIP cosine similarity of at least 0.98.

Retrieval used CLIP ViT-L/14-336 image embeddings [22]. The 27 nearest eligible images were shuffled deterministically by question identifier into 9 groups of 3. In the shared condition, every agent received the first group; in the private condition, agent i received group i. For each retrieved image, the question–answer pair with maximal term frequency–inverse document frequency (TF-IDF) cosine similarity to the current question was selected. Examples with images supplied that pair alongside its image. Text-only examples supplied the same question and answer without the image. The single agent also received shared examples with images or as text.

S1.5. Query-agnostic reports and coverage

Query-agnostic reports separated perception from reasoning [21]. GPT-5 mini, Gemini 3.7 Flash and Claude Haiku 4.5 each served as a perception module and a reasoning module. Perception modules received neither the question nor its answer options. They were instructed to write at most 120 words that described modality and plane alongside visible anatomy, and to record every abnormal or notable finding with its location.

Each perception module generated 9 reports per image or multi-image set, reused across questions about that image. All reasoning modules received identical reports for a given perception module, image and report index. Reasoning agents received the report, question and options without the image. Each agent used its own report in the own-report condition. All agents used the first report in the shared-report condition, which used matching perception and reasoning modules. Comparisons with direct image input used self-consistency samples from the same reasoning model.

Report coverage on SLAKE used a synonym list based on the supplied organ and lesion labels. A report covered the queried target when it mentioned the target or a listed synonym. Accuracy and error correlation were evaluated by report condition, and errors were separated by target coverage.

S1.6. Open-weight answer probabilities

Open-weight models used one forward pass per item and condition without sampling. They received the same rendered pixels as the proprietary models. Softmax normalized next-token logits over the answer words yes/no for binary items and over option letters for multiple-option items. Each answer used the highest logit among accepted capitalization and leading-space token variants. The validation comparison that fixed the yes/no readout appears in Tab. S1.

Visual reliance compared original-image and blank-image probability distributions. On yes/no items it was the absolute change in the log-odds of yes. On MedXpertQA-MM it was the Kullback–Leibler divergence from the original-image distribution to the blank-image distribution. Error correlation between models was also divided by its maximum given their marginal error rates, $\phi_{\max}$. Spearman correlation related shared wrong answers of proprietary models to open-weight visual reliance.

S1.7. Matched contrasts of error correlation

The primary component analysis used matched differences in mean pairwise error correlation. Each contrast subtracted a condition in which agents shared the named component from one in which they did not:

- **Role:** Independent MAS agents with subspecialty roles minus self-consistency samples with the same general-radiologist prompt.
- **Contrast adjustment:** agents with different contrast adjustments minus self-consistency samples with original images.
- **In-context examples:** agents with private examples minus agents with shared examples, matched by example format, with images or as text.
- **Report:** the own-report condition minus the shared-report condition, with matching perception and reasoning modules.
- **Perception module:** agents that read reports from different perception modules minus agents that read different reports from one perception module, within a reasoning model.
- **Reasoning module:** agents from different reasoning models that read the same report minus agents from one reasoning model that read one shared report.
- **Model:** self-consistency samples from different models minus self-consistency samples from one model.

Contrasts were computed within each reasoning model or model pair and then averaged over models or model pairs. For the reasoning-module and model contrasts, the subtracted mean averaged the separate within-model correlations of both models. Cross-model comparisons reused the existing self-consistency responses. Self-consistency error correlation from one model was the reference; it was reported as ϕ, rather than $\Delta\phi$ (Tab. S8).

S1.8. Regression and statistical inference

Ordinary least squares regression supplied an additive secondary summary of pairwise error correlation (Tab. S9). Predictors indicated components that two agents did not share, including image versus report input. The regression included one intercept per pair of reasoning models. Its coefficients were conditional on the other predictors; the matched contrasts were the primary component analysis.

Error correlation was the mean Pearson correlation of binary error indicators over agent pairs. Pairs with constant error indicators were omitted. For yes/no systems, the beta-binomial majority prediction used mean agent accuracy and error correlation; a majority required at least 5 correct agents out of 9. Shared wrong answers required at least 7 of 9 agents to be wrong. Conformity was the share of initially correct agents whose final answer was wrong [1, 33]. It was stratified by initial majority status and by agreement with the same agent's blank-image answer.

Unparsed answers counted as wrong. These abstentions were reported by condition and mostly reflected refusals to read blank images. A sensitivity analysis recomputed error correlation on items that both agents answered. Pair accuracy required both answers to be correct, with answers compared as option text, yes or no.

S2. Validation before the main experiments

Validation fixed generation settings and the open-weight probability readout before the main experiments (Tab. S1). Accuracy differences in this table use pp.

Table S1. **Validation fixed the settings used in the controlled evaluation.** V1–V6 cover answer parsing, image preparation, retrieval, localization, reports and open-weight readout. CLAHE denotes contrast-limited adaptive histogram equalization; CLIP denotes Contrastive Language–Image Pretraining; CT denotes computed tomography.

Check	Result	Setting fixed
V1	Single-agent answer parsing was 100% for all four proprietary models called through OpenRouter. On the same prompt, GPT-5 mini used 256 reasoning tokens at low effort and 2,304 at high effort.	GPT-5 mini used low reasoning effort.
V2	Single-agent accuracy relative to the original image over 10 model-by-data-set cells: CLAHE averaged 0.0 pp (range −6 to 14); grayscale inversion averaged −6.6 pp (range −12 to 2). The changes at 256 and 128 pixels were −1.0 and −5.6 pp.	Inversion exceeded the 5 pp limit. Contrast adjustment used gamma 0.7 for agents 4–6 and CLAHE for agents 7–9.
V3	Parsing was 100% with four images per prompt: the query and three in-context examples. A CLIP threshold of 0.95 removed about 32 similar cases from different patients per query. Radiographs from different patients reached cosines of 0.91 to 0.95. A perceptual hash missed the same CT level and the same patient's chest radiograph (cosine 0.99).	In-context examples used the hash and similarity exclusion rules in Sec. S1, with CLIP threshold 0.98.
V4	In 20 inspected SLAKE-target cases, the target box covered the queried organ or lesion. The control box did not overlap the target and lay within the body.	Target and control boxes were used as built.
V5	Query-agnostic reports had median lengths of 72 to 77 words and a maximum of 109 words. Answer parsing was 100%.	The report protocol was used as built.
V6	With option-letter probabilities, the LLaVA family assigned 96–100% probability to "A" regardless of the image. Pooled accuracy on 100 items, 50 each from VQA-RAD and SLAKE: MedGemma 84%, HuatuoGPT-Vision 79%, Qwen3-VL 78%, Qwen2-VL 74%, LLaVA-1.5 55% and LLaVA-Med 53%.	Binary items used yes/no token logits. LLaVA-Med was excluded by the 55% pooled-accuracy threshold.
Runtime	Qwen 3.8 Flash used 250 to 6,400 reasoning tokens and 6 to 77 seconds per call.	Qwen 3.8 Flash used original, reduced-resolution and blank images, self-consistency and contrast adjustment.

S3. Data collection

All proprietary-model calls used OpenRouter in October 2026. Perception modules generated reports before reasoning-module calls. GPT-5 mini and Claude Haiku 4.5 each had a median report length of 75 words; Gemini 3.7 Flash had a median of 90. Maximum lengths were 118, 121 and 127 words, respectively. Provider routing and token budgets remained fixed. Transient service failures were retried until every item had a response. The automatic stop at more than 15% unparsed answers was not applied to blank-image conditions.

S4. Additional results

S4.1. Nine agents are worth fewer than two

Multi-agent gains differed mainly in the accuracy of the agents before voting. For GPT-5 mini, Gemini 3.7 Flash and Claude Haiku 4.5, Independent gains ranged from -2.8 to 4.0 pp and Decentralized gains from -3.2 to 5.6 pp. Claude's Decentralized MAS gained 5.6 pp on VQA-RAD. Qwen 3.8 Flash's Independent MAS gained 13.6 pp on VQA-RAD, 8.0 pp on SLAKE and 5.6 pp on MedXpertQA-MM; its Decentralized gains ranged from 8.8 to 14.0 pp. For the other models, majority voting changed accuracy by -2.1 to 2.4 pp relative to mean agent accuracy (Tab. S2).

Debate produced unanimity even on incorrectly answered items. Qwen's final-answer ϕ ranged from 0.822 to 0.913. Final answers were unanimous on 82.4% to 100.0% of items, and 8.9% to 70.0% of unanimous answers were wrong (Tab. S2).

The beta-binomial fit held across systems with different error correlations. The largest absolute residual was 2.59 pp; majority minus mean agent accuracy ranged from -2.1 to 5.6 pp over $\phi = 0.44$ to 0.99 (Fig. S1).

S4.2. Shared wrong answers follow the language prior

Blank-image agreement also identified shared errors on the binary-question data sets. Shared wrong answers matched the blank-image majority in 42.9% to 54.4% of cases on VQA-RAD and 14.3% to 57.9% on SLAKE (Tab. S4).

Single-agent blank-image losses differed across data sets. They ranged from 11.2 to 26.8 pp on VQA-RAD and 25.2 to 40.0 pp on SLAKE. On MedXpertQA-MM, losses were 4.0 pp for GPT-5 mini, 3.6 pp for Claude and 2.0 pp for Qwen; Gemini lost 18.0 pp (Tab. S3).

Blank images raised error correlation. For Gemini, Claude and Qwen, blank-image ϕ ranged from 0.841 to 0.922 on VQA-RAD and 0.831 to 0.982 on SLAKE, compared with original-image ranges of 0.439 to 0.731 and 0.529 to 0.785, respectively (Tab. S3).

The rise in error correlation held on items both agents answered. Claude's blank-image ϕ on items both agents answered was 0.963 on VQA-RAD, 0.931 on SLAKE and 0.852 on MedXpertQA-MM (Tab. S3).

Reduced resolution moved error correlation in opposite directions across models. On VQA-RAD, GPT-5 mini's ϕ fell from 0.752 to 0.644, while Qwen's rose from 0.439 to 0.480 (Tab. S3 and Fig. S2).

Answer cross-tabulation separated visual benefits from answers that became correct without the image. On VQA-RAD, 24.1% to 37.3% of agent answers were correct only with the original image, versus 10.4% to 13.6% only with the blank image. SLAKE shares were 36.3% to 46.4% versus 4.4% to 15.8%; MedXpertQA-MM shares were 8.1% to 22.3% versus 4.4% to 7.9%. On MedXpertQA-MM, 12.3% to 64.0% were wrong with both images (Tab. S4).

Open-weight models also lost accuracy without visual evidence. Blank images cost four of five models 16.4 to 19.2 pp on VQA-RAD; LLaVA-1.5 7B lost 2.0 pp. All five lost 16.0 to 48.8 pp on SLAKE. On MedXpertQA-MM, original-image accuracy was 20.0% to 24.4%, versus 20.0% to 23.6% with blank images (Tab. S11 and Fig. S3).

Open-weight probability changes showed visual reliance beyond answer changes. Mean absolute change in the log-odds of yes ranged from 0.89 to 6.43 on VQA-RAD and 1.40 to 11.23 on SLAKE. On MedXpertQA-MM, Kullback–Leibler divergence ranged from 0.10 to 0.88 (Tab. S11 and Fig. S3).

Proprietary-model shared wrong answers were associated with lower open-weight visual reliance on SLAKE. Spearman ρ was -0.401 on SLAKE, -0.048 on VQA-RAD and 0.006 on MedXpertQA-MM (Fig. S3).

S4.3. Models differ in where they look

Occlusion effects differed across proprietary and open-weight models. GPT-5 mini's sensitivity was 17.1 pp and Claude's was 3.2 pp. Open-weight estimates ranged from 2.4 to 21.6 pp; LLaVA-1.5 7B had 2.4 pp (Tab. S5).

Models sometimes gave identical answers to images with opposite reference answers. Gemini's pair accuracy was 74.0%. Single-agent answers differed within 75.0% of pairs for Gemini, 62.0% for GPT-5 mini and 42.5% for Claude; remaining pairs received identical answers (Tab. S5).

S4.4. Perception sets accuracy; a shared report sets correlation

Gemini reports improved the other reasoning modules' accuracy relative to image input. The endpoint effects on VQA-RAD were 6.1 pp for GPT-5 mini and 16.3 pp for Claude. Across reasoning modules, Gemini reports yielded 78.7% to 81.3% accuracy on VQA-RAD and 84.3% to 88.2% on SLAKE (Tab. S7).

Claude reports reduced the other reasoning modules' accuracy relative to image input. They lowered accuracy by 7.9 to 25.9 pp in 5 of 6 cells. The remaining cell, GPT-5 mini on SLAKE, differed by -0.4 pp (Tab. S7).

Reasoning modules also differed when they read GPT-5 mini reports. On MedXpertQA-MM, accuracy was 60.8% for Gemini, 50.1% for GPT-5 mini and 30.6% for Claude (Tab. S7).

Query-agnostic reports omitted queried targets even when they improved overall accuracy. The target was absent from 26.5% to 30.8% of SLAKE reports and from 25.8% to 47.1% of reports behind errors. For Gemini reports, omissions occurred in 26.7% overall and 33.6% to 47.1% of reports behind errors across reasoning modules (Tab. S7).

S4.5. Changing the model decorrelates errors most

Changes to shared components produced different reductions in error correlation. Contrast adjustment lowered ϕ by 0.045 to 0.094; private in-context examples lowered it by 0.110 to 0.146 relative to shared examples. Own reports lowered ϕ by 0.172 to 0.204 relative to a shared report. Changing the perception module lowered it by 0.263 to 0.311; reasoning-module contrasts under identical reports ranged from $\Delta\phi = -0.430$ to -0.108 (Tab. S8).

Private examples lowered error correlation and accuracy together. For Claude on SLAKE, private image examples lowered ϕ from 0.928 to 0.655 while MAS accuracy fell from 70.4% to 68.8% (Tab. S10).

The additive regression supplied a conditional summary alongside the matched contrasts. Its coefficients summarize the components agents did not share after adjustment for other predictors and a separate intercept for each reasoning-model pair (Tab. S9).

Open-weight models sharing an encoder did not have the highest normalized error correlation. Between-model ϕ ranged from 0.099 to 0.535. HuatuoGPT-Vision 7B and LLaVA-1.5 7B share the CLIP ViT-L/14-336 encoder; their $\phi/\phi_{\max}$ ranged from 0.213 to 0.451, versus 0.148 to 0.725 for pairs with different encoders (Tab. S11 and Fig. S3).

S4.6. Debate removes correct minority answers

Conformity was higher on MedXpertQA-MM when correct answers differed from the agent's blank-image answer. Paired differences ranged from 4.6 pp for Gemini to 17.4 pp for Claude. Among correct answers that differed from the blank-image answer, 6.6% to 23.5% were initially in the minority (Tab. S6).

S5. Detailed results

Detailed results cover original-image MAS (Tab. S2), reduced resolution and blank images (Tab. S3), blank-image answer agreement (Tab. S4), region occlusion and paired images (Tab. S5), conformity (Tab. S6), query-agnostic reports (Tab. S7), matched contrasts (Tab. S8), additive regression (Tab. S9), in-context examples (Tab. S10) and open-weight models (Tab. S11).

Table S2. Multi-agent systems of 9 agents with original images (250 items per data set): accuracy (%), multi-agent gain (pp), its split into mean agent minus single agent and majority minus mean agent, error correlation ϕ of initial and final answers, N_{eff}, and the final answers of the Decentralized MAS that are unanimous and, among those, wrong (%).

Data set	Model	Single agent	Mean agent	Indep.	Decentr.	Indep. − single	Decentr. − single	Mean agent − single	Majority − mean agent	ϕ initial	N_{eff}	ϕ final	Unanimous	Wrong if unanimous
VQA-RAD	GPT-5 mini	72.0	71.3	69.2	70.0	−2.8	−2.0	−0.7	−2.1	0.752	1.28	1.000	100.0	30.0
	Gemini 3.7 Flash	81.2	81.3	82.4	81.2	1.2	0.0	0.1	1.1	0.725	1.32	0.997	99.6	18.9
	Claude Haiku 4.5	60.8	65.9	64.8	66.4	4.0	5.6	5.1	−1.1	0.731	1.31	0.996	99.2	33.5
	Qwen 3.8 Flash	69.6	79.9	83.2	80.4	13.6	10.8	10.3	3.3	0.439	1.99	0.822	86.0	15.8
SLAKE	GPT-5 mini	80.8	79.8	78.4	80.8	−2.4	0.0	−1.0	−1.4	0.785	1.24	0.995	99.6	18.9
	Gemini 3.7 Flash	91.2	89.9	90.0	89.2	−1.2	−2.0	−1.3	0.1	0.785	1.24	1.000	100.0	10.8
	Claude Haiku 4.5	72.4	72.3	72.4	74.4	0.0	2.0	−0.1	0.1	0.775	1.25	1.000	100.0	25.6
	Qwen 3.8 Flash	76.4	83.3	84.4	90.4	8.0	14.0	6.9	1.1	0.529	1.72	0.884	94.4	8.9
MedXpertQA-MM	GPT-5 mini	54.8	53.2	55.6	58.4	0.8	3.6	−1.6	2.4	0.685	1.39	0.996	98.8	41.7
	Gemini 3.7 Flash	84.4	83.3	83.6	82.4	−0.8	−2.0	−1.1	0.3	0.749	1.29	1.000	100.0	17.6
	Claude Haiku 4.5	33.2	30.8	30.4	30.0	−2.8	−3.2	−2.4	−0.4	0.836	1.17	1.000	100.0	70.0
	Qwen 3.8 Flash	43.6	46.7	49.2	52.4	5.6	8.8	3.1	2.5	0.647	1.46	0.913	82.4	42.7

Table S3. Reduced resolution and blank images: single-agent accuracy (%) and error correlation ϕ of the Independent MAS of 9 agents with original images ($\leq$1024 pixels), images capped at 256 and 128 pixels, and blank images (uniform gray, same size); single-agent accuracy lost with the blank image (pp); unparsed answers with the blank image (%) and ϕ recomputed on items both agents answered.

Data set	Model	Acc. original	Acc. 256 px	Acc. 128 px	Acc. blank	ϕ original	ϕ 256 px	ϕ 128 px	ϕ blank	Original − blank	Unparsed, blank	ϕ blank, answered
VQA-RAD	GPT-5 mini	72.0	74.0	72.8	60.8	0.752	0.694	0.644	0.742	11.2	0.0	–
	Gemini 3.7 Flash	81.2	83.6	77.6	54.4	0.725	0.726	0.749	0.922	26.8	0.0	–
	Claude Haiku 4.5	60.8	63.2	65.2	46.8	0.731	0.706	0.591	0.876	14.0	10.4	0.963
	Qwen 3.8 Flash	69.6	67.6	72.0	54.8	0.439	0.442	0.480	0.841	14.8	0.1	–
SLAKE	GPT-5 mini	80.8	78.8	75.2	54.4	0.785	0.792	0.709	0.802	26.4	0.0	–
	Gemini 3.7 Flash	91.2	91.2	85.2	51.2	0.785	0.803	0.820	0.982	40.0	0.0	–
	Claude Haiku 4.5	72.4	73.6	70.0	38.8	0.775	0.752	0.718	0.879	33.6	4.3	0.931
	Qwen 3.8 Flash	76.4	74.8	79.2	51.2	0.529	0.547	0.490	0.831	25.2	0.0	–
MedXpertQA-MM	GPT-5 mini	54.8	50.4	50.8	50.8	0.685	0.690	0.700	0.730	4.0	0.0	–
	Gemini 3.7 Flash	84.4	80.4	78.4	66.4	0.749	0.762	0.784	0.799	18.0	0.0	–
	Claude Haiku 4.5	33.2	32.0	32.0	29.6	0.836	0.821	0.840	0.818	3.6	14.8	0.852
	Qwen 3.8 Flash	43.6	40.8	43.2	41.6	0.647	0.630	0.609	0.656	2.0	0.5	–

Table S4. Agent answers of the Independent MAS of 9 cross-tabulated by correctness with the original and the blank image (% of answers), and shared wrong answers (at least 7 of 9 agents wrong) whose majority answer matches the blank-image majority.

Data set	Model	Right with both	Right only with original	Wrong only with original	Wrong with both	Shared wrong answers (n)	Matching blank-image majority (%)
VQA-RAD	GPT-5 mini	47.2	24.1	12.0	16.7	57	54.4
	Gemini 3.7 Flash	44.0	37.3	10.4	8.3	35	42.9
	Claude Haiku 4.5	36.6	29.3	13.6	20.5	70	54.3
	Qwen 3.8 Flash	46.0	33.9	10.8	9.3	18	44.4
SLAKE	GPT-5 mini	43.5	36.3	9.5	10.7	42	57.1
	Gemini 3.7 Flash	46.7	43.2	4.4	5.8	19	57.9
	Claude Haiku 4.5	25.9	46.4	15.8	11.9	56	33.9
	Qwen 3.8 Flash	38.5	44.8	12.0	4.7	21	14.3
MedXpertQA-MM	GPT-5 mini	39.7	13.5	7.9	38.9	95	64.2
	Gemini 3.7 Flash	61.0	22.3	4.4	12.3	31	74.2
	Claude Haiku 4.5	22.7	8.1	5.2	64.0	163	71.2
	Qwen 3.8 Flash	32.9	13.8	7.0	46.4	105	76.2

Table S5. Region occlusion (SLAKE-target, 250 items) and paired images (SLAKE-pairs, 200 pairs). Answer-change rates under target and control occlusion (%; proprietary models: agents of the Independent MAS of 9; open-weight models: one forward pass), occlusion sensitivity (target minus control, pp), pair accuracy and the share of pairs answered differently (%; single agent).

Model		Target change	Control change	Occlusion sensitivity	Pair accuracy	Answers differ
GPT-5 mini	proprietary	23.5	6.4	17.1	59.0	62.0
Gemini 3.7 Flash	proprietary	36.4	3.6	32.9	74.0	75.0
Claude Haiku 4.5	proprietary	10.7	7.5	3.2	38.5	42.5
MedGemma 4B	open-weight	13.2	1.6	11.6	62.5	63.5
Qwen3-VL 8B	open-weight	23.6	3.6	20.0	53.5	56.5
Qwen2-VL 7B	open-weight	11.2	4.8	6.4	49.5	53.5
HuatuoGPT-Vision 7B	open-weight	28.0	6.4	21.6	48.5	53.0
LLaVA-1.5 7B	open-weight	12.8	10.4	2.4	31.5	39.5

Table S6. Conformity in the Decentralized MAS of 9 agents: share of correct initial answers that are wrong after debate (%), for answers in the initial minority or majority, and for answers that differ from or match the same agent's blank-image answer (difference in pp); share of correct initial answers in the initial minority among those that differ from or match the blank-image answer (%).

Data set	Model	Initial minority	Initial majority	Differs from blank	Matches blank	Difference	In minority, differs	In minority, matches
VQA-RAD	GPT-5 mini	72.3	1.9	9.8	4.6	5.1	11.2	3.8
	Gemini 3.7 Flash	100.0	1.0	3.7	3.4	0.3	2.5	2.6
	Claude Haiku 4.5	75.0	1.4	7.1	5.7	1.4	8.5	5.3
	Qwen 3.8 Flash	54.1	6.5	12.5	6.6	5.9	6.2	4.9
SLAKE	GPT-5 mini	56.1	1.1	4.5	1.9	2.6	4.0	3.4
	Gemini 3.7 Flash	71.4	1.2	2.9	1.4	1.5	1.8	1.0
	Claude Haiku 4.5	62.5	1.4	4.0	3.4	0.6	3.4	4.8
	Qwen 3.8 Flash	41.5	1.8	4.2	2.8	1.4	3.4	5.5
MedXpertQA-MM	GPT-5 mini	68.5	1.4	18.1	2.7	15.4	20.1	3.5
	Gemini 3.7 Flash	71.7	1.9	7.2	2.6	4.6	6.6	1.5
	Claude Haiku 4.5	83.9	4.4	23.6	6.3	17.4	19.8	3.9
	Qwen 3.8 Flash	62.2	2.4	18.1	5.4	12.7	23.5	6.2

Table S7. Perception and reasoning. (a) Mean agent accuracy (%) of each reasoning module when every agent reads its own query-agnostic report from the named perception module, or reads the image (self-consistency samples of the reasoning module); the change from the image (pp); error correlation ϕ of 9 agents reading the image, their own reports and one shared report (perception and reasoning module the same model). (b) SLAKE: reports that do not mention the queried target, among all reports and among the reports behind wrong answers (%).

(a) Accuracy and error correlation

Data set	Reasoning module	Reports by GPT-5 mini	Reports by Gemini	Reports by Claude	Image	GPT-5 mini reports − image	Gemini reports − image	Claude reports − image	ϕ image	ϕ own reports	ϕ shared report
VQA-RAD	GPT-5 mini	69.0	78.7	61.6	72.6	−3.6	6.1	−10.9	0.741	0.818	0.944
	Gemini 3.7 Flash	70.2	81.3	63.6	80.3	−10.1	1.0	−16.7	0.776	0.783	0.990
	Claude Haiku 4.5	70.6	78.8	63.5	62.5	8.1	16.3	0.9	0.878	0.723	0.986
SLAKE	GPT-5 mini	83.5	87.4	78.8	79.1	4.4	8.3	−0.4	0.800	0.777	0.955
	Gemini 3.7 Flash	84.9	88.2	80.2	89.7	−4.8	−1.6	−9.5	0.828	0.771	0.965
	Claude Haiku 4.5	81.0	84.3	76.3	73.4	7.6	10.9	2.9	0.882	0.728	0.969
MedXpertQA-MM	GPT-5 mini	50.1	62.0	45.0	52.9	−2.8	9.1	−7.9	0.692	0.641	0.800
	Gemini 3.7 Flash	60.8	74.5	57.8	83.7	−22.9	−9.2	−25.9	0.811	0.715	0.880
	Claude Haiku 4.5	30.6	45.6	27.4	31.7	−1.1	13.8	−4.4	0.915	0.701	0.893

(b) Target absent from the report, SLAKE

Reasoning module	GPT-5 mini reports, all	GPT-5 mini reports, behind errors	Gemini reports, all	Gemini reports, behind errors	Claude reports, all	Claude reports, behind errors
GPT-5 mini	26.5	29.8	26.7	36.4	30.8	40.6
Gemini 3.7 Flash	26.5	25.8	26.7	33.6	30.8	39.1
Claude Haiku 4.5	26.5	29.9	26.7	47.1	30.8	42.9

Table S8. Matched contrasts of error correlation: ϕ of two self-consistency agents of one model, and the mean change $\Delta\phi$ over models (or model pairs) when two agents do not share the named component. In-context examples were run on VQA-RAD and SLAKE.

	VQA-RAD	SLAKE	MedXpertQA-MM
Self-consistency ϕ, GPT-5 mini	0.741	0.800	0.692
Self-consistency ϕ, Gemini 3.7 Flash	0.776	0.828	0.811
Self-consistency ϕ, Claude Haiku 4.5	0.878	0.882	0.915
Self-consistency ϕ, Qwen 3.8 Flash	0.442	0.604	0.629
$\Delta\phi$, Role	−0.048	−0.060	−0.033
$\Delta\phi$, Contrast adjustment	−0.092	−0.094	−0.045
$\Delta\phi$, In-context examples	−0.110	−0.146	–
$\Delta\phi$, Report	−0.199	−0.204	−0.172
$\Delta\phi$, Perception module	−0.311	−0.275	−0.263
$\Delta\phi$, Reasoning module	−0.108	−0.167	−0.430
$\Delta\phi$, Model	−0.470	−0.473	−0.492

Table S9. Additive regression of pairwise error correlation on the components two agents do not share (one intercept per pair of reasoning modules), a secondary summary of Tab. S8.

Data set	Perception module	Image vs report	Contrast adjustment	Role	In-context examples	Report	Reasoning module
VQA-RAD	−0.247	−0.123	−0.023	−0.006	−0.026	0.150	−0.026
SLAKE	−0.193	−0.094	−0.017	−0.004	−0.040	0.009	−0.124
MedXpertQA-MM	−0.160	−0.099	−0.012	−0.009	–	−0.066	−0.194

Table S10. In-context examples (VQA-RAD and SLAKE): single-agent accuracy, accuracy of the Independent MAS of 9 agents (%) and its error correlation ϕ without examples, with shared examples (the same 3 for every agent) and with private examples (different examples per agent), given with their images or as text. The single agent received shared examples only.

Data set	Model		None	Shared, images	Private, images	Shared, text	Private, text
VQA-RAD	GPT-5 mini	Single agent	72.0	73.6	–	75.6	–
		MAS accuracy	73.6	74.8	75.2	76.4	74.8
		ϕ	0.741	0.751	0.668	0.737	0.703
	Gemini 3.7 Flash	Single agent	81.2	82.4	–	83.6	–
		MAS accuracy	80.4	82.8	82.0	84.4	84.4
		ϕ	0.776	0.708	0.673	0.743	0.692
	Claude Haiku 4.5	Single agent	60.8	68.8	–	61.2	–
		MAS accuracy	62.0	65.2	64.4	62.4	62.4
		ϕ	0.878	0.890	0.664	0.912	0.681
SLAKE	GPT-5 mini	Single agent	80.8	79.2	–	81.6	–
		MAS accuracy	78.4	81.6	84.0	82.0	81.6
		ϕ	0.800	0.741	0.667	0.780	0.690
	Gemini 3.7 Flash	Single agent	91.2	89.2	–	90.8	–
		MAS accuracy	90.0	89.2	90.4	90.4	92.0
		ϕ	0.828	0.836	0.722	0.823	0.716
	Claude Haiku 4.5	Single agent	72.4	67.6	–	76.8	–
		MAS accuracy	73.6	70.4	68.8	78.0	78.0
		ϕ	0.882	0.928	0.655	0.941	0.720

Table S11. Open-weight models (one forward pass per item). (a) Accuracy (%) with original, blank, 256-pixel and 128-pixel images, and mean visual reliance ($|\Delta$ log-odds of yes$|$ between original and blank image on yes/no items; Kullback–Leibler divergence on MedXpertQA-MM). (b) Error correlation ϕ between two models and $\phi/\phi_{\max}$, its ratio to the maximum given the two marginal error rates; HuatuoGPT-Vision 7B and LLaVA-1.5 7B share the CLIP ViT-L/14-336 encoder.

(a) Accuracy and visual reliance

Model	Data set	Original	Blank	256 px	128 px	Visual reliance
MedGemma 4B	VQA-RAD	80.0	60.8	80.4	76.0	6.43
	SLAKE	88.8	40.0	86.8	84.0	11.23
	MedXpertQA-MM	22.0	23.2	22.4	22.4	0.88
Qwen3-VL 8B	VQA-RAD	74.0	57.6	70.0	68.8	5.98
	SLAKE	79.2	43.6	83.6	81.2	9.72
	MedXpertQA-MM	24.4	23.6	25.6	25.2	0.72
Qwen2-VL 7B	VQA-RAD	76.8	58.8	73.6	76.4	1.49
	SLAKE	78.0	43.2	72.8	68.0	2.44
	MedXpertQA-MM	20.8	21.6	21.2	21.2	0.11
HuatuoGPT-Vision 7B	VQA-RAD	74.8	56.4	77.2	75.6	0.89
	SLAKE	76.8	50.8	76.0	76.0	1.84
	MedXpertQA-MM	20.0	20.0	19.6	20.4	0.10
LLaVA-1.5 7B	VQA-RAD	56.0	54.0	56.4	56.0	1.10
	SLAKE	66.0	50.0	66.4	66.0	1.40
	MedXpertQA-MM	21.2	20.4	21.2	20.8	0.10

(b) Error correlation between models

Model pair	Shared encoder	VQA-RAD, ϕ	VQA-RAD, $\phi/\phi_{\max}$	SLAKE, ϕ	SLAKE, $\phi/\phi_{\max}$	MedXpertQA-MM, ϕ	MedXpertQA-MM, $\phi/\phi_{\max}$
MedGemma 4B – Qwen3-VL 8B		0.365	0.432	0.224	0.324	0.238	0.254
MedGemma 4B – Qwen2-VL 7B		0.294	0.323	0.485	0.725	0.227	0.236
MedGemma 4B – HuatuoGPT-Vision 7B		0.332	0.385	0.316	0.488	0.338	0.359
MedGemma 4B – LLaVA-1.5 7B		0.181	0.321	0.227	0.459	0.244	0.250
Qwen3-VL 8B – Qwen2-VL 7B		0.322	0.348	0.418	0.433	0.282	0.313
Qwen3-VL 8B – HuatuoGPT-Vision 7B		0.412	0.421	0.535	0.574	0.275	0.312
Qwen3-VL 8B – LLaVA-1.5 7B		0.099	0.148	0.360	0.505	0.161	0.176
Qwen2-VL 7B – HuatuoGPT-Vision 7B		0.227	0.239	0.371	0.385	0.507	0.520
Qwen2-VL 7B – LLaVA-1.5 7B		0.162	0.261	0.373	0.504	0.192	0.195
HuatuoGPT-Vision 7B – LLaVA-1.5 7B	yes	0.172	0.263	0.346	0.451	0.205	0.213

S6. Supplementary figures

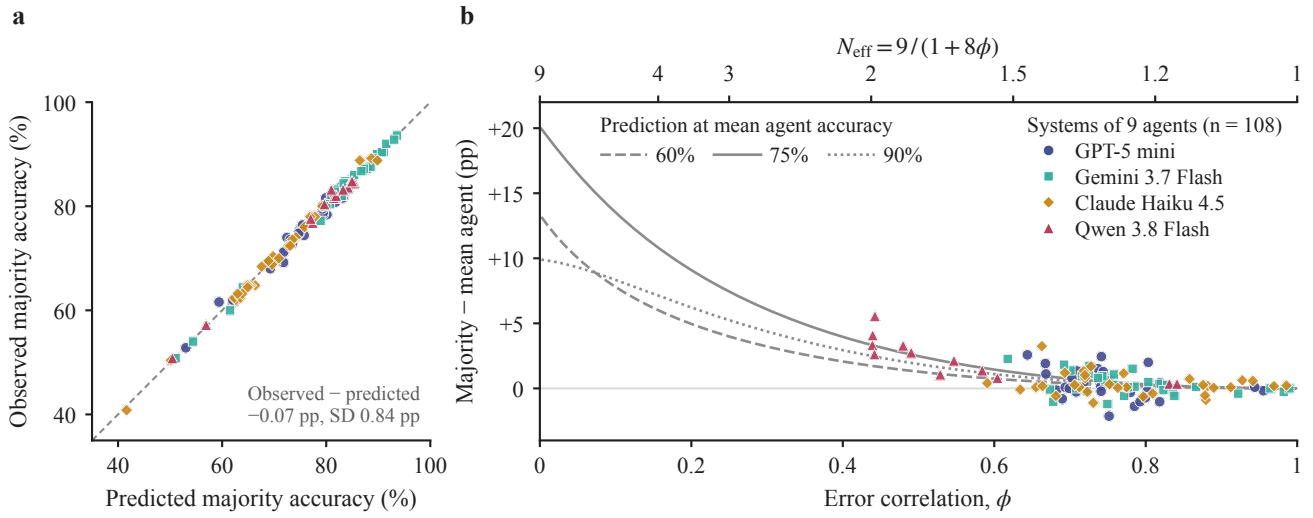


Figure S1. **Mean agent accuracy and error correlation predict majority accuracy on yes/no items.** (a) Observed versus beta-binomial predicted majority accuracy for 108 systems of 9 agents; the dashed diagonal marks equality. Observed minus predicted accuracy averaged −0.07 pp, with standard deviation (SD) 0.84 pp. (b) Majority minus mean agent accuracy against error correlation ϕ; gray curves give predictions at mean agent accuracy 60% (dashed), 75% (solid) and 90% (dotted). The top axis gives $N_{\text{eff}} = 9/(1+8\phi)$. The systems used VQA-RAD, SLAKE and SLAKE-target (250 items each), and SLAKE-pairs (200 pairs). Model colors as in Fig. 1.

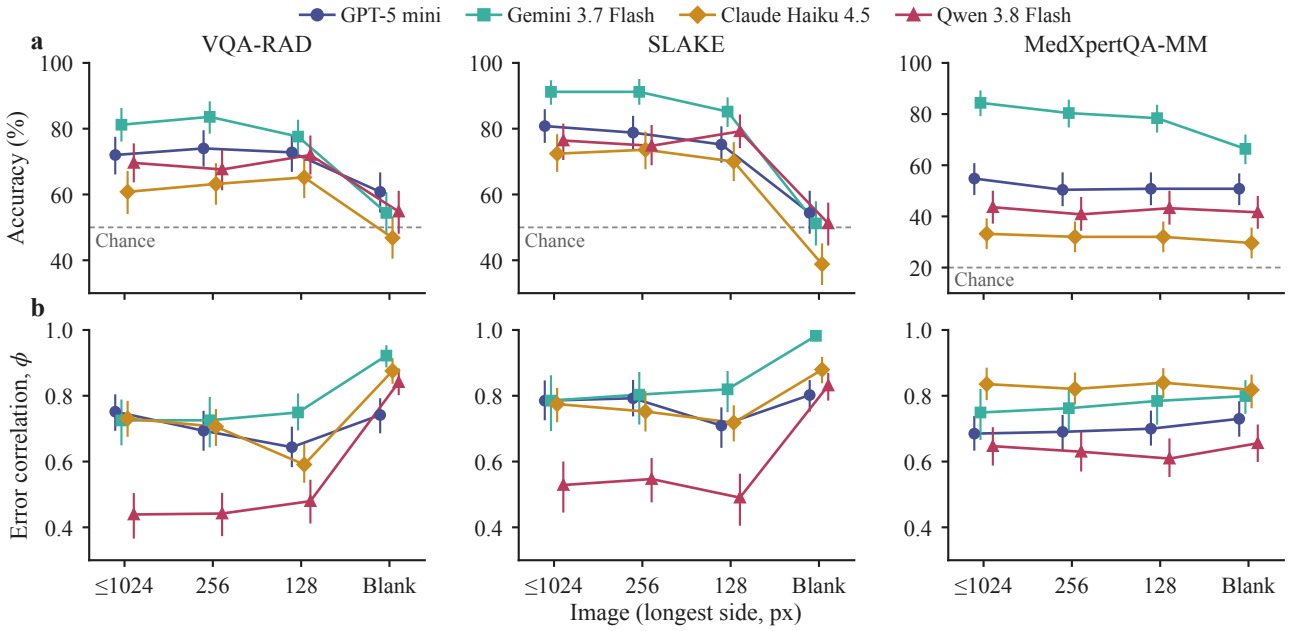


Figure S2. **Reduced resolution changes error correlation in different directions across models.** (a) Single-agent accuracy and (b) error correlation ϕ of the Independent MAS across image conditions. Columns show VQA-RAD, SLAKE and MedXpertQA-MM, with 250 items each. Labels $\leq$ 1024, 256 and 128 cap the longest image side in pixels; Blank denotes a uniform gray image of unchanged dimensions. Lines join adjacent conditions. Dashed lines mark chance accuracy. Model colors as in Fig. 1.

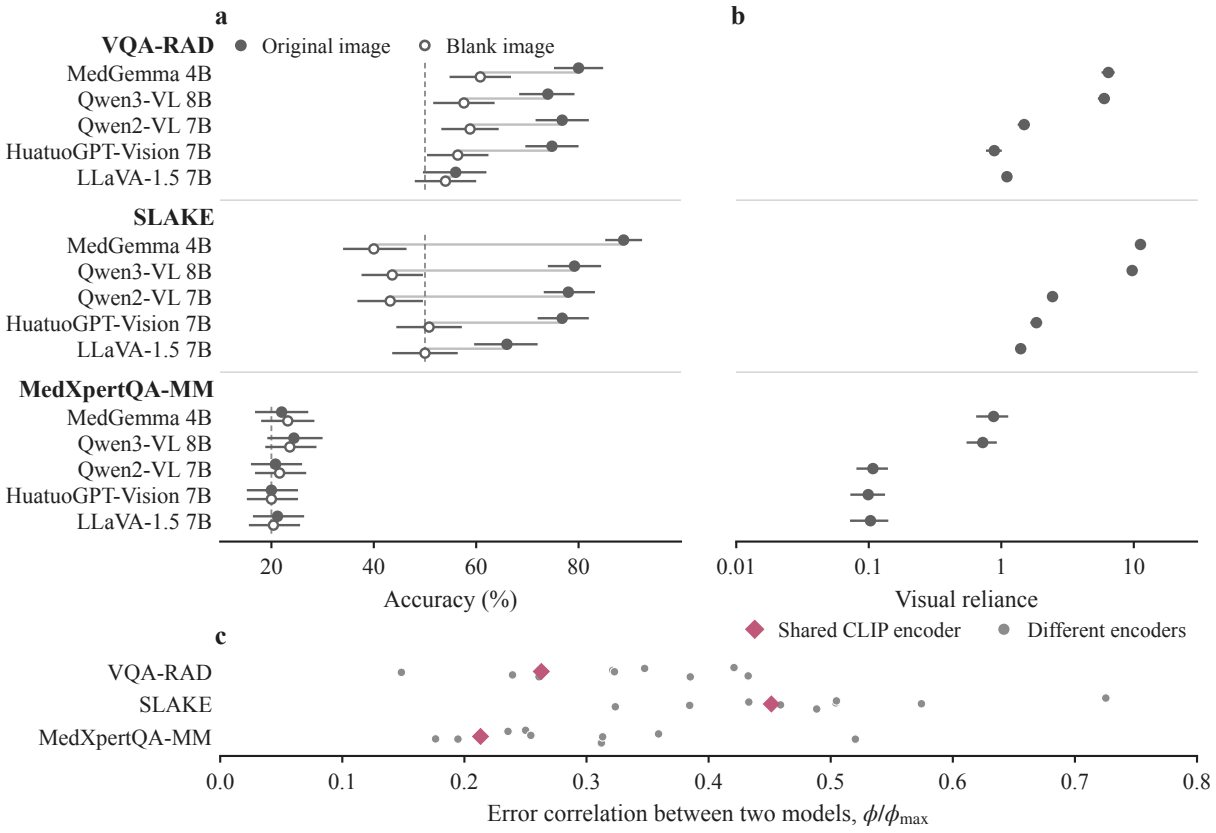


Figure S3. **Open-weight probes show visual reliance without sampling.** Each model used one forward pass per item on VQA-RAD, SLAKE and MedXpertQA-MM, with 250 items each. (a) Accuracy with the original image (filled) and blank image (open); dashed vertical lines mark chance. (b) Visual reliance on a log scale: absolute change in log-odds of yes on binary items, and Kullback–Leibler divergence from original-image to blank-image probabilities on MedXpertQA-MM. Points are means. Dark gray markers in (a,b) identify the labeled open-weight models. (c) Between-model error correlation divided by its maximum given the marginal error rates, $\phi/\phi_{\max}$. Rose diamonds identify HuatuoGPT-Vision 7B and LLaVA-1.5 7B, which share the CLIP ViT-L/14-336 encoder; gray circles identify pairs with different encoders.

S7. Text-only clinical questions

An unpublished companion analysis evaluated language-model agents on text-only clinical questions. The data sets were MedQA, MedAgentsBench and the text set of MedXpertQA [10, 25, 36]. The 8 models were Claude Haiku 4.5, Claude Sonnet 5, DeepSeek V4 Flash, Gemini 3.5 Flash Lite, Gemini 3.7 Flash, GPT-4.1 nano, GPT-5 mini and GPT-5 nano. They answered as single agents and in multi-agent systems with specialty roles. The 340 configurations comprised 96 Independent, 96 Centralized, 96 Decentralized and 52 Hybrid systems, with architecture names from Kim et al. [14].

Initial-answer error correlation in the companion analysis was within the imaging range. Across the 288 Independent, Centralized and Decentralized configurations with 3 to 9 agents, mean initial-answer ϕ was 0.758. These architectures shared initial answers and had identical initial-answer correlations by design. With 9 agents, ϕ was 0.762 and N_{eff} was 1.27.

Debate increased error correlation in the companion Decentralized systems. Across the 96 configurations, mean ϕ was 0.758 (SD 0.117) before debate and 0.990 (SD 0.010) after debate. After debate, N_{eff} was 1.01.